

This system uses dictionaries to store detailed position information, lists to track purchase dates and summary information, and dictionary methods to safely look up current prices.

Market Screener

This example uses sets and dictionaries to filter stocks based on multiple criteria:

```
1. # Stock universe with key metrics
2. stock_universe = {
3.     "AAPL": {"pe_ratio": 28.5, "market_cap": 2400, "sector": "Technology", "dividend_yield":
4.     0.6},
5.     "JNJ": {"pe_ratio": 15.2, "market_cap": 430, "sector": "Healthcare", "dividend_yield": 2.7},
6.     "XOM": {"pe_ratio": 10.1, "market_cap": 380, "sector": "Energy", "dividend_yield": 3.8},
7.     "PG": {"pe_ratio": 25.3, "market_cap": 350, "sector": "Consumer Staples", "dividend_yield":
8.     2.4},
9.     "JPM": {"pe_ratio": 14.2, "market_cap": 420, "sector": "Financials", "dividend_yield": 2.5}
10. }
11.
12. # Filter criteria
13. value_stocks = set()
14. growth_stocks = set()
15. income_stocks = set()
16.
17. for symbol, metrics in stock_universe.items():
18.     # Value criteria: low P/E ratio
19.     if metrics["pe_ratio"] < 15:
20.         value_stocks.add(symbol)
21.
22.     # Growth criteria: technology sector or high P/E
23.     if metrics["sector"] == "Technology" or metrics["pe_ratio"] > 25:
24.         growth_stocks.add(symbol)
25.
26.     # Income criteria: dividend yield over 2%
27.     if metrics["dividend_yield"] > 2.0:
28.         income_stocks.add(symbol)
29.
30. # Stocks that meet multiple criteria
31. value_and_income = value_stocks.intersection(income_stocks)
```